EXP 7

Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits. Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

AIM

To create a Servlet-based web application that stores user preferences (such as theme or language) using cookies and retrieves them on subsequent visits.

ALGORITHM

1. Start the application.
2. Accept user preferences (theme/language) through a form.
3. Create cookies for each preference.
4. Set the cookie expiry time.
5. Store cookies in the client’s browser.
6. When the user revisits the application:
    - Read cookies from the request.
    - Check for stored preferences.
7. If cookies are found:
    - Retrieve and display the stored preferences.
8. If cookies are not found:
    - Display default values or ask the user to set preferences.
9. Stop the application.

STEPS

1. Design a user interface to collect preferences like theme and language.
2. Configure a Servlet to handle form submission.
3. Use cookies to store the user’s preferences in the browser.
4. Set an appropriate expiry time for cookies.
5. Send cookies along with the response to the client.
6. On subsequent visits, retrieve cookies from the request object.
7. Extract stored values (theme/language).
8. Display the preferences to the user dynamically.

9. Handle cases where cookies are absent by showing default settings.

Program:

DisplayServlet.java:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class DisplayServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String theme = "Light";
        String language = "English";
        Cookie[] cookies = request.getCookies();

        if (cookies != null) {
            for (Cookie c : cookies) {
                if (c.getName().equals("theme")) {
                    theme = c.getValue();
                }
                if (c.getName().equals("language")) {
                    language = c.getValue();
                }
            }
        }

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
```

```
String bgColor = theme.equals("Dark") ? "#1e1e2f" : "#f0f4f8";
String cardColor = theme.equals("Dark") ? "#2c2c3c" : "#ffffff";
String textColor = theme.equals("Dark") ? "#ffffff" : "#2c3e50";
out.println("<html>");
out.println("<head>");
out.println("<title>Select a book</title>");

out.println("<style>");
out.println("body { font-family: 'Segoe UI', sans-serif; margin:0; padding:0; text-align:center; }");

out.println(".card { margin:60px auto; padding:30px; width:420px; border-radius:12px; box-shadow:0px 5px 15px rgba(0,0,0,0.1); }");

out.println("h2 { margin-bottom:10px; }");

out.println("p { font-size:16px; }");

out.println(".notice { margin-top:20px; font-size:18px; font-weight:bold; }");

out.println(".btn { display:inline-block; margin-top:25px; padding:10px 15px; background:#3498db; color:white; text-decoration:none; border-radius:6px; }");

out.println(".btn:hover { background:#2980b9; }");

out.println("</style>");
out.println("</head>");

// Body start
out.println("<body style='background:" + bgColor + "; color:" + textColor + ";'>");

out.println("<div class='card' style='background:" + cardColor + ";'>");
```

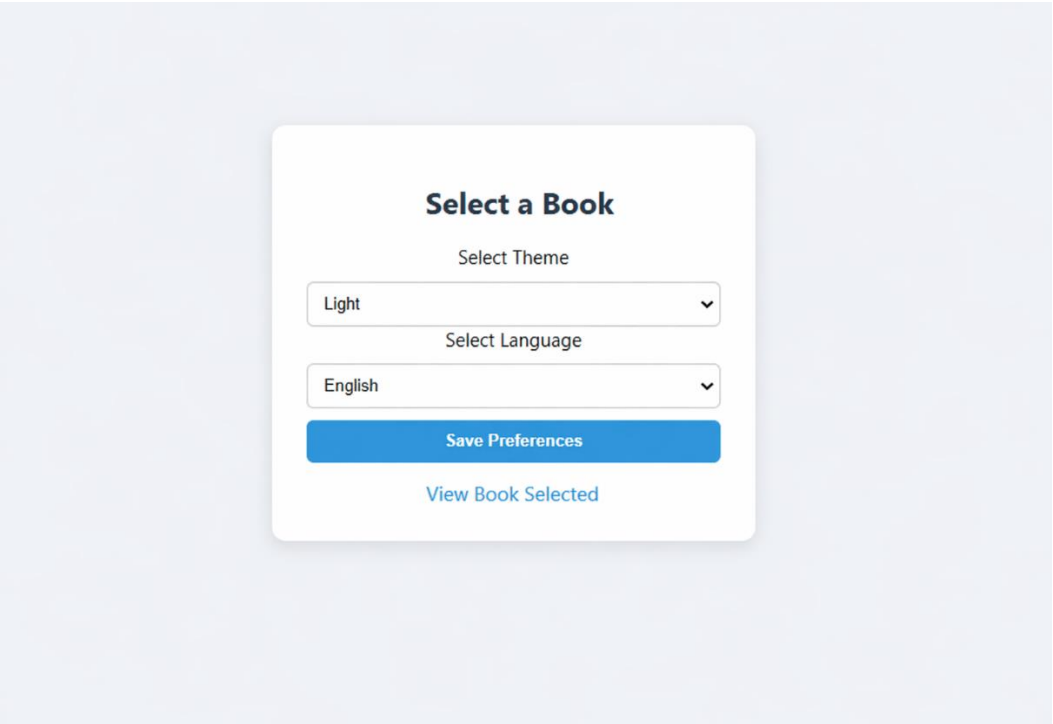
```
        out.println("<h2>Library System</h2>");
        out.println("<h3>Welcome Back!</h3>");

        out.println("<p><b>Theme:</b> " + theme + "</p>");
        out.println("<p><b>Language:</b> " + language + "</p>");

        // Notice based on language
        if (language.equals("Tamil")) {
            out.println("<div class='notice'>Tamil book</div>");
        } else {
            out.println("<div class='notice'>English book</div>");
        }

        out.println("<a class='btn' href='index.html'>Change Preferences</a>");

        out.println("</div>");
        out.println("</body>");
        out.println("</html>");
    }
}
```

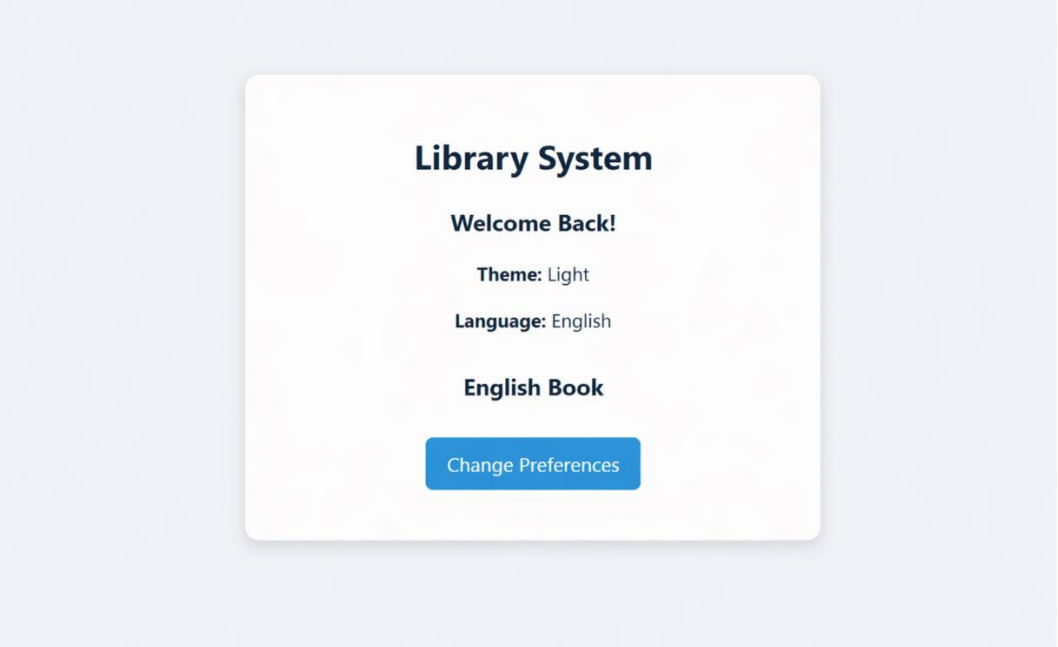
PreferenceServlet.java:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class PreferenceServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
```

```
        String theme = request.getParameter("theme");
        String language = request.getParameter("language");

        // Create cookies
        Cookie themeCookie = new Cookie("theme", theme);
        Cookie langCookie = new Cookie("language", language);

        // Set expiry (1 day)
        themeCookie.setMaxAge(60 * 60 * 24);
        langCookie.setMaxAge(60 * 60 * 24);

        // Add cookies to response
        response.addCookie(themeCookie);
        response.addCookie(langCookie);

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h3>Preferences Saved Successfully!</h3>");
        out.println("<a href='DisplayServlet'>Go to Select a book</a>");
    }
}
```

OUTPUT:





RESULT

The Servlet application successfully stores user preferences using cookies and retrieves them on subsequent visits, providing a personalized user experience.